Ahmedabad
University

CSE623 Machine Learning

Weekly Report 7

Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis

Submitted to faculty: Prof. Mehul Raval

Date of Submission: 04-04-2026

Student Details

Roll No.	Name of the Student
AU2544013	Hardi Makwana
AU2544022	Lakshita Rathod
AU2544028	Siddhi Hirani
AU2544030	Krisha Doshi

2026-2027 (Winter Semester)

Aim:

The first main aim of this week was to create a classical Machine Learning baseline using the Support Vector Machines to conduct pixel-wise semantic segmentation of wheat spikes from images taken from complex scenarios in the fields. The task was to find the most discriminative color and textural characteristics that make the target class of "Spike" distinguishable against other classes such as leaves, stems, and soil.

Introduction:

Segmentation of wheat spikes is an important task for phenotyping and estimating yield. While deep learning techniques use automated feature extraction, our approach uses hand-crafted feature extraction along with a Linear Support Vector Classifier (LinearSVC). The advantages of this technique include its high interpretability due to the possibility to see what visual features (for instance, yellow color of spike or directionality of a leaf) are the most discriminative.

Work Completed:

A) Post-Processing and Instance Segmentation

Because classical machine learning models evaluate data strictly on a pixel-by-pixel basis, the raw predictive output often suffers from localized spatial noise and the inability to distinguish between physically touching objects. To transition the Random forest model's output from raw semantic masks into accurate, quantifiable instance counts, a two-stage mathematical post-processing pipeline was implemented specifically to refine the predictions generated by the advanced **16-feature** and **25-feature** models.

I) Morphological Noise Reduction

Due to the fact that classical machine learning algorithms consider each pixel individually when assessing an image, the resulting output can be highly noisy and fail to identify separate objects that may touch one another. In order to convert the predictions made by the Random Forest model from their semantic mask form into actual quantifiable instances, a two-step mathematical process was employed, which was uniquely designed to analyze predictions from both complex models.

- **Morphological Opening (Erosion followed by Dilation):** The process helped remove stray pixels with low density from the background without altering the size of actual wheat targets.
- **Morphological Closing (Dilation followed by Erosion):** Carried out right after opening, this process joined internal spaces along with smoothing external outlines of the predicted wheat kernels to form a solid shape.

Results :

- For 16 Features

```
Train IoU: 0.3846 | Validation IoU: 0.2788
```

- For 25 Features

```
Train IoU: 0.4144 | Validation IoU: 0.3039
```

B) Advanced Feature Engineering

We implemented a robust feature extraction engine that generates a multi-dimensional descriptor for every pixel. The most advanced version (v7) utilizes **14 specialized features** categorized into three groups:

- **Color Context (5 features):** LAB B-channel (Yellow-Blue balance), Excess Green-Red Index (ExGR), and Normalized Green-Red Difference Index (NGRDI).
- **Texture (5 features):** Uniform LBP, to describe micro texture patterns; Gabor Isotropy (to account for the multi-directional properties of wheat awns); and local entropy.
- **Spatial Context (4 features):** Color and edge-density pooling, like ExGR_pool15 and Sobel_pool5, respectively, which endow each pixel with information on its surrounding area, thus greatly reducing the amount of noise.

C) Model Architecture & Training Strategy

To overcome the computational intensity of traditional SVMs, we utilized:

- **LinearSVC:** A fast, linear solver optimized for large pixel datasets.
- **CalibratedClassifierCV:** A wrapper using **Platt Scaling** to transform raw SVM decision scores into probabilities, allowing for fine-tuned thresholding.
- **Hard Negative Mining:** A two-stage training process where the model is first trained, then re-educated specifically on the leaf and stem pixels it initially misclassified as spikes.

D) Performance Metrics (v7 Results)

The integration of spatial pooling and hard negative mining in the final SVM version yielded the following results:

Metric	Training Pool	Local Validation (Image-level)
Mean IoU (Spike)	0.5512	0.1706

Mean F1-Score	**0.5100**	**0.2292**
Pixel Accuracy	**~67%**	**N/A**

Next steps and goals:

A. Feature Space Optimization

Rather than training and testing additional models for 7, 10, 16, and 25 features, going forward the pipeline will use solely the most effective set of features moving forward. This will be done by utilizing our past knowledge of feature importance to filter out unneeded features, thereby limiting the number of features to those that have maximum discrimination capability.

B. Transition to Random Forest

Although SVM gave us a good starting point, we move on to Random Forest (RF). Random Forest is better suited for:

- **Handling Imbalanced Data:** The ensemble approach and bagging in RF make it easier to deal with the very large majority class, "Background," compared to the linear model of SVM.
- **Non-Linear Relationships:** Without needing the extra work of crafting kernels for an SVM, RF can discover intricate relationships between features.

C. Watershed Algorithm for Instance Segmentation

To shift focus from semantic segmentation to individual counting, we introduce the Watershed Algorithm. Using watershed on the distance map generated by our existing segmentation boundaries, we expect to:

- Properly split clumped or merged wheat heads.
- Count the heads accurately on a per-instance basis.

Conclusion:

The move from a traditional **LinearSVC** to **Random Forests** combined with **Watershed-based instance segmentation** demonstrates a clear change from a mere classification approach to the development of a sophisticated biologically-aware phenotyping tool. In contrast to the success demonstrated by the SVM baseline with its **IoU of 0.5512** in training, the implementation of an ensemble-based model helps to mitigate the natural shortcomings associated with linear classifiers.

In this regard, the inclusion of **morphological refinement** and **distance transform segmentation** allows establishing an efficient way of processing the information derived from semantic noise to the point when biological measurements can be taken into account.

References:

1. D. P. David et al., "Global Wheat Head Dataset: A Large and Diverse Dataset of High-Resolution RGB Images to Develop and Benchmark Wheat Head Detection Methods," https://www.sciencedirect.com/science/article/pii/S2643651524000359

2. M. Kass, A. Witkin, and D. Terzopoulos, "Snakes: Active Contour Models," https://web.cs.ucla.edu/~dt/papers/ijcv88/ijcv88.pdf

3. J. Ji et al., "Comparison of Different Machine Learning Algorithms for the Prediction of the Wheat Grain Filling Stage Using RGB Images," https://www.mdpi.com/2223-7747/12/23/4043

4. S. S. Sannakki, V. S. Rajpurohit, V. B. Nargund, and P. Kulkarni, "Extraction of Texture Features Using Gabor Filter for Wheat Crop Disease Detection" https://ijret.org/volumes/2015v04/i12/IJRET20150412043.pdf

5. Y. Guo et al., "Otsu Multi-Threshold Image Segmentation Based on Adaptive Double-Mutation Differential Evolution" https://www.mdpi.com/2313-7673/8/5/418

6. A. Farhan, M. A. Alqaralleh, and H. Alwazneh, "Texture Based Leaf Disease Classification Using Machine Learning Techniques". https://www.researchgate.net/publication/364096456_Texture_based_Leaf_Disease_classification_using_Machine_Learning_Techniques

7. A. R. Pathak, M. Pandey, and S. Rautaray, "Plant Disease Diagnosis Using Image Processing Techniques – A Review on Machine and Deep Learning Approaches," https://www.mdpi.com/2079-9292/11/17/2641